LearnEase AI – AI-Powered Course Content Simplification Agent

Domain: Education – Artificial Intelligence (AI)

Problem Statement

Challenge: College students often struggle to understand complex academic concepts and lengthy study materials. Preparing simplified notes, examples, and exam questions also takes considerable time.

Objective: To develop an AI-powered application that converts complex study material into simple, structured, and exam-oriented learning content.

Proposed Features:

- Simplifies complex topics into beginner-friendly explanations.
- Generates definitions, key points, and important terms.
- Provides real-life examples.
- Creates exam-oriented questions and answers.
- Generates concise revision summaries.
- Accepts topics, text, PDF, and DOCX files as input.

Technology: Python, Streamlit, IBM watsonx.ai Runtime, and Llama 3.3 70B Instruct.

Future Scope: Add multilingual support, personalized learning, and advanced AI workflow integration.